

WT : EXPERIMENT 5

Aim : Simulation of Wireless notice board using tinkercad

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Theory :

The concept of a Wireless Notice Board involves the use of embedded systems and wireless communication technologies to create a system that can display messages or notices on a board without the need for physical connections. We are focusing on a Wireless Notice Board system that will use Arduino and Bluetooth communication for transmitting and receiving messages remotely.

The system will utilize a Bluetooth module (such as the HC-05), an Arduino Uno, and a 16x2 LCD display to create a working prototype of a wireless notice board. Additionally, we will simulate this setup using Tinkercad, an online simulation platform, where the components and their connections can be modeled, and the code can be tested.

Working Principle:

The **Wireless Notice Board** system operates as follows:

1. Bluetooth Communication:

- The user can send a message from a smartphone using a Bluetooth terminal app (like **Bluetooth Terminal**). The HC-05 Bluetooth module on the Arduino receives the message wirelessly.
- Once the Bluetooth module (HC-05) receives the data, it passes the message to the Arduino Uno via serial communication. The Arduino processes the data and then sends it to the LCD display.

2. Displaying the Message on the LCD:

- The Arduino takes the message received from the Bluetooth module and sends it to the 16x2 LCD display. The LiquidCrystal library is used to control the LCD, and the message is printed on the screen in real-time.
- If the message exceeds the length of the LCD (16 characters per row), the system can either scroll the text or display the message in multiple parts, ensuring that all the information is shown to the user.

3. Real-Time Updates:

- The system continuously monitors the Bluetooth module for incoming data. When a new message is received, it is immediately displayed on the notice board (LCD), making it suitable for real-time information updates.

Steps for Simulation:

1. Build the Circuit:

1. Place the Components:

- Arduino Uno: Place the Arduino Uno board in the Tinkercad workspace.
- TMP36 Temperature Sensor: Drag the TMP36 temperature sensor onto the workspace.
 - Connect the VCC pin of the TMP36 to the 5V pin on the Arduino.
 - Connect the GND pin of the TMP36 to the GND pin on the Arduino.
 - Connect the OUT pin of the TMP36 to A0 (analog pin) on the Arduino.
This will allow the Arduino to read the temperature as an analog voltage.
- 16x2 LCD Display: Drag the 16x2 LCD display to the workspace.
 - Connect the VCC pin of the LCD to the 5V pin on the Arduino.
 - Connect the GND pin of the LCD to the GND pin on the Arduino.
 - Connect the SDA pin of the LCD to A4 (SDA pin on Arduino).
 - Connect the SCL pin of the LCD to A5 (SCL pin on Arduino).
- Jumper Wires: Use jumper wires to make the connections on the breadboard or directly between the components and the Arduino.

2. Upload the Code

3. Test the System:

Once the circuit is set up and the code is uploaded to the Arduino, we can simulate the system in Tinkercad.

1. Simulate the Circuit:

- Click on **Start Simulation** in Tinkercad to begin the process.
- The LCD will display the initial message and then continuously update with the current temperature read by the TMP36 sensor.

2. Adjust the Temperature:

- If we want to simulate different temperatures, we can change the sensor voltage by adjusting the input voltage to the TMP36 sensor to simulate different temperature readings.

Code :

```
#include <LiquidCrystal.h>
```

```
// Pin for the temperature sensor
```

```
const int tempPin = A0; // TMP36 sensor connected to A0 pin
```

```
LiquidCrystal lcd_1(12, 11, 5, 4, 3, 2);
```

```
void setup()
```

```
{
```

```
  lcd_1.begin(16, 2); // Set up the number of columns and rows on the LCD.
```

```
  // Print a welcome message to the LCD.
```

```
  lcd_1.print("Temperature:");
```

```
}
```

```
void loop()
```

```
{
```

```
  // Read the temperature sensor (analog value between 0 and 1023)
```

```
  int sensorValue = analogRead(tempPin);
```

```
  // Convert the sensor value to voltage (0-5V range)
```

```
  float voltage = sensorValue * (5.0 / 1024.0); // Convert analog value to voltage
```

```
  // Convert the voltage to temperature (in Celsius) for TMP36
```

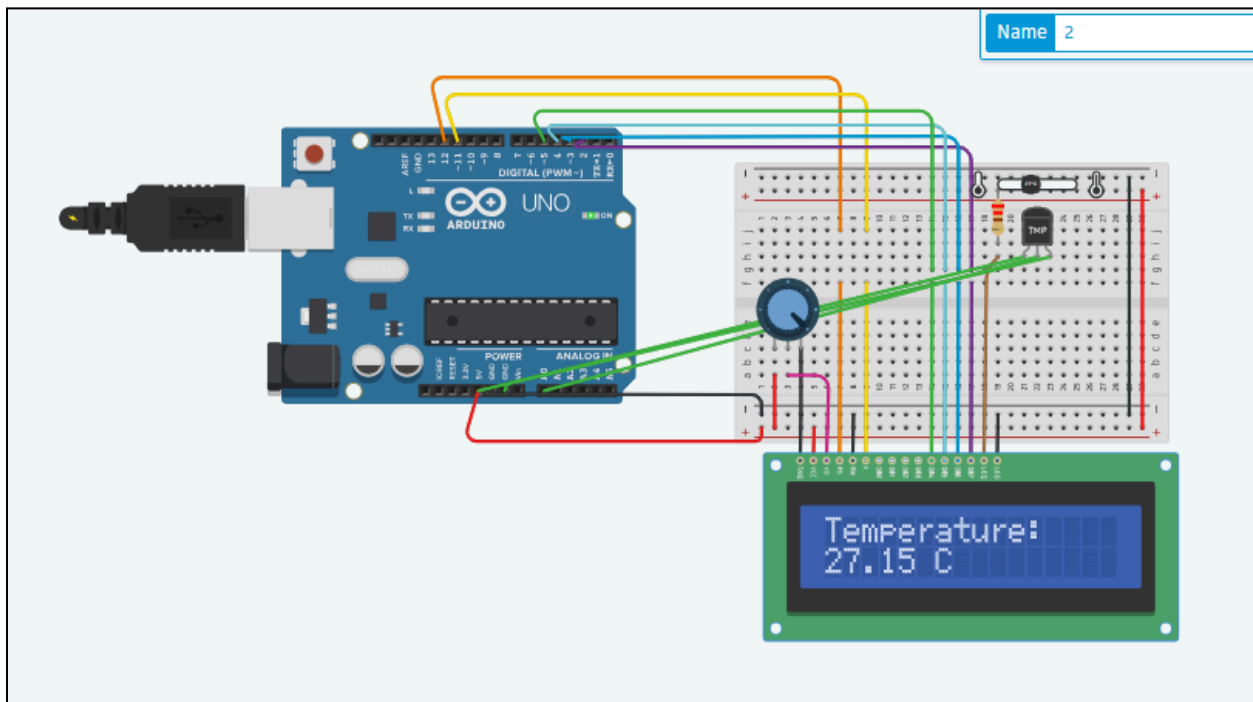
```
// TMP36 has 500mV offset at 0°C, and each degree Celsius increases the voltage by 20mV  
float temperatureC = (voltage - 0.5) / 0.01;
```

```
// Set the cursor to the first row, second column (0,1)  
lcd_1.setCursor(0, 1);
```

```
// Print the temperature value to the LCD  
lcd_1.print(temperatureC); // Display temperature in Celsius  
lcd_1.print(" C "); // Display unit (Celsius) and some padding
```

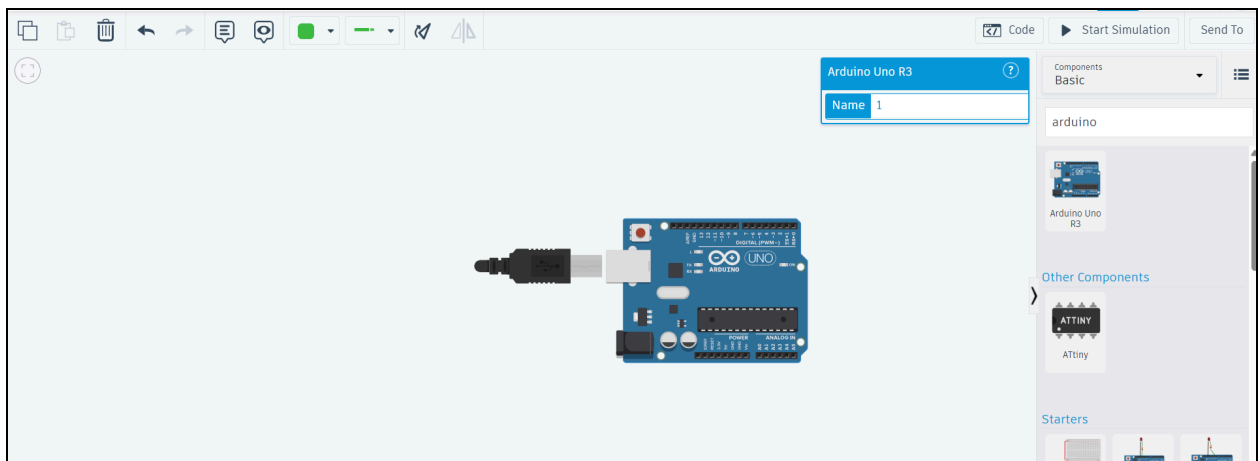
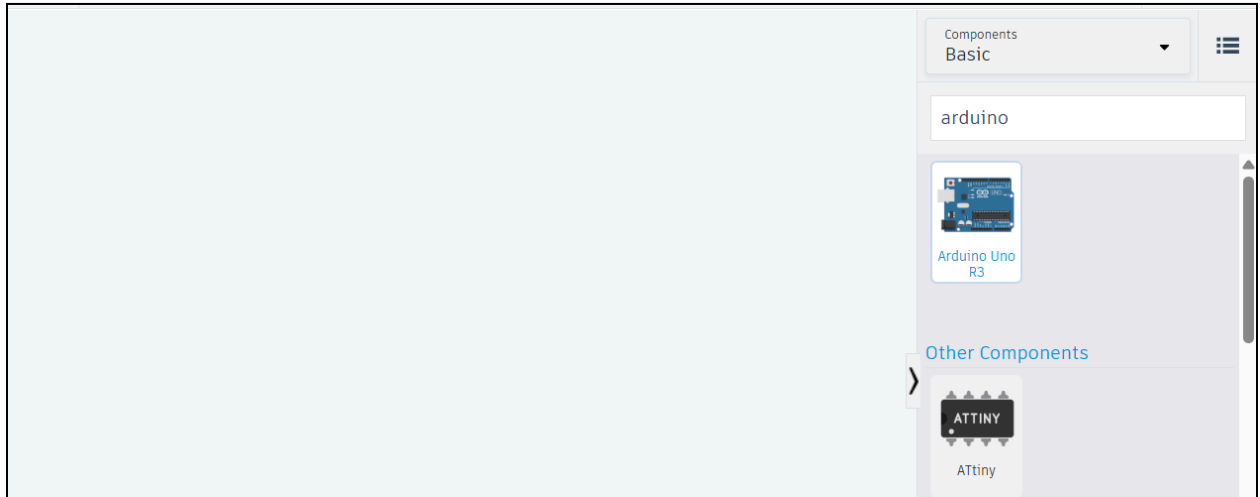
```
// Wait for a short time before updating the temperature reading  
delay(2000); // Update every 2 seconds  
}
```

Output :

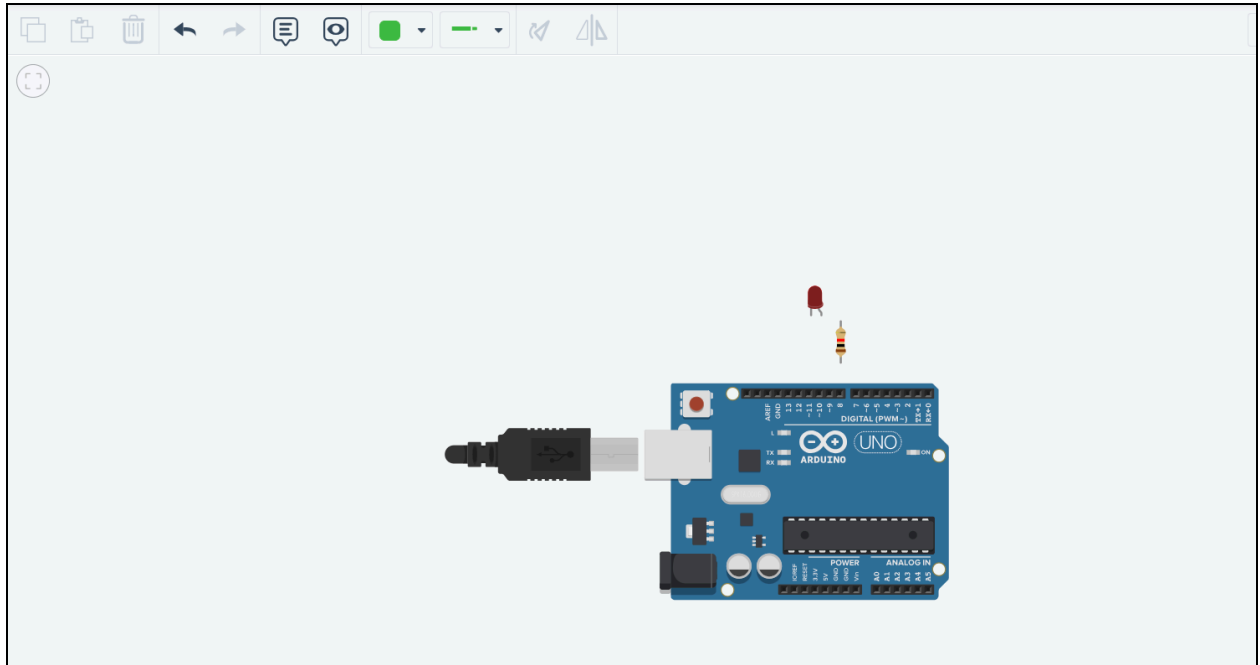


Tinkercad - Blinking LED

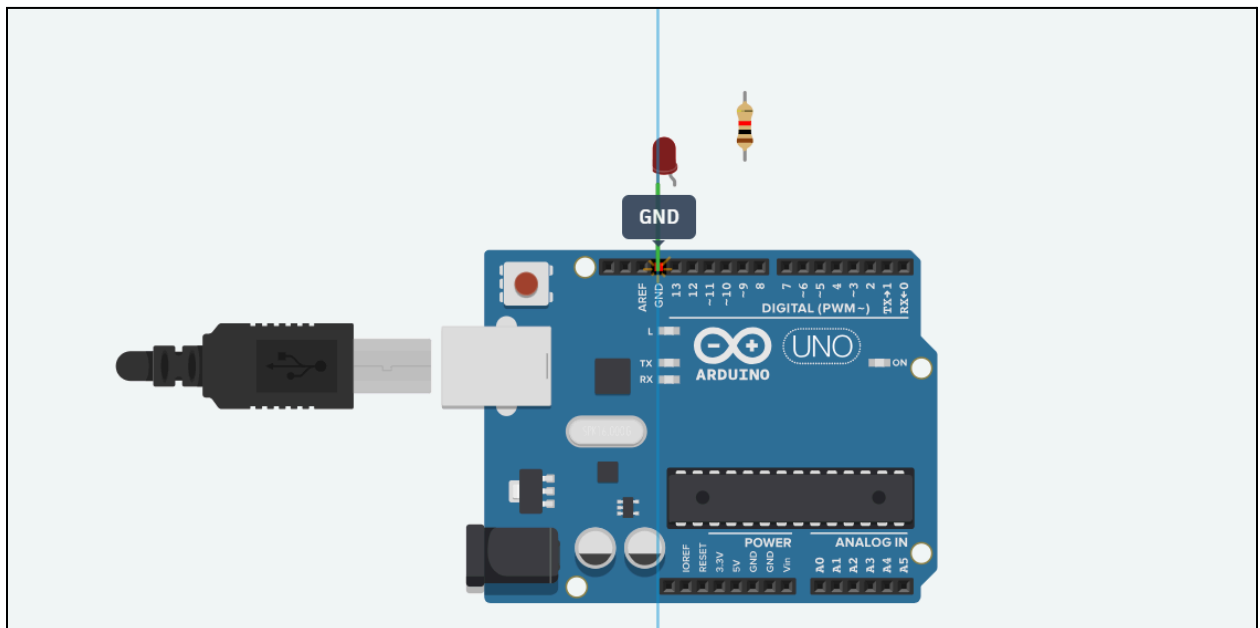
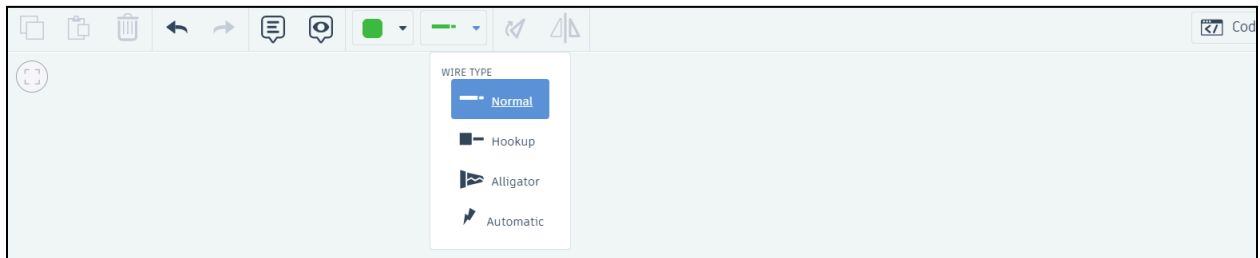
Search for components in the sidebar and drag and drop the selected component in the white area which is the workspace.

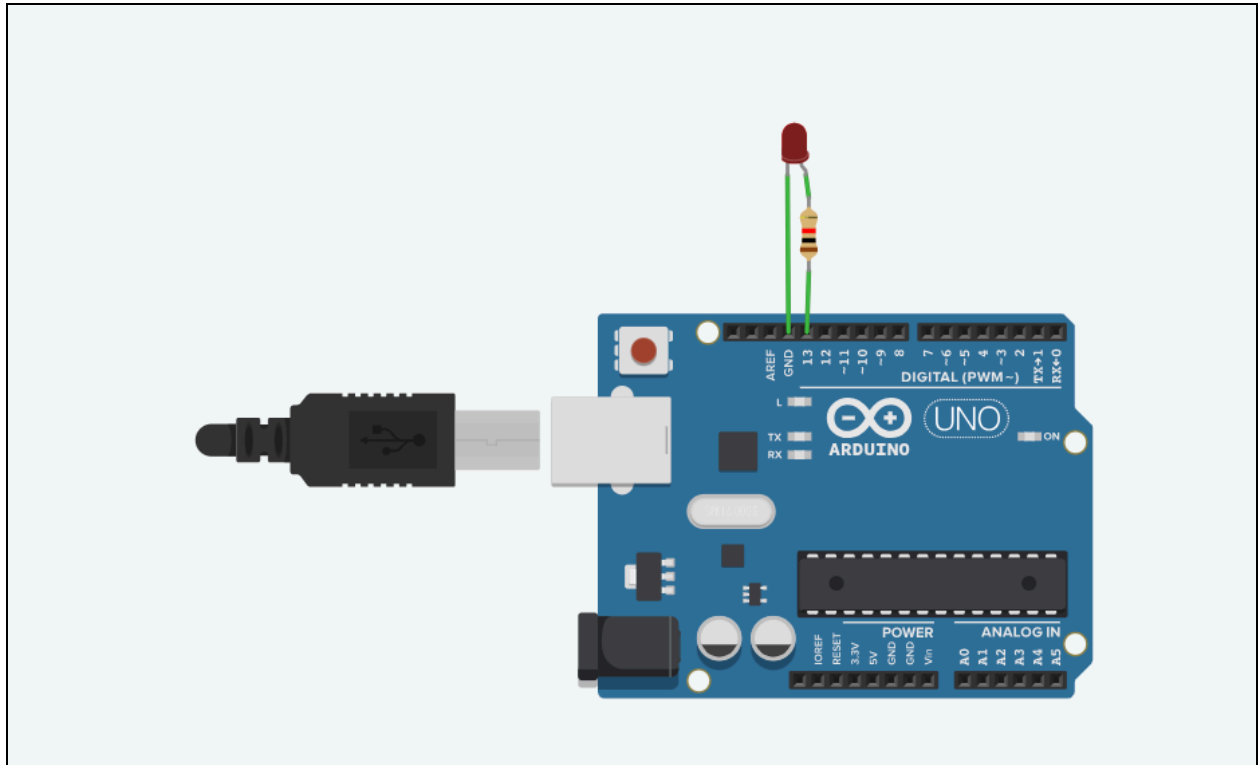


For blinking LED, we need 3 components. Arduino, LED and resistor. Drag all the components in the workplace.

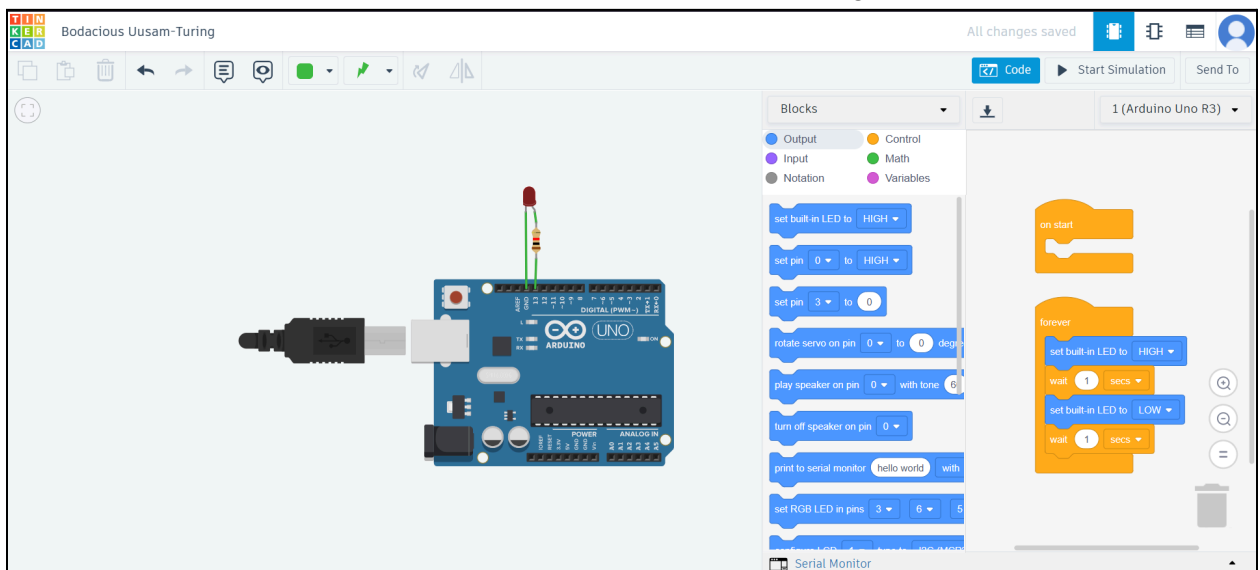


Connect the components with the help of wires available in the top bar.

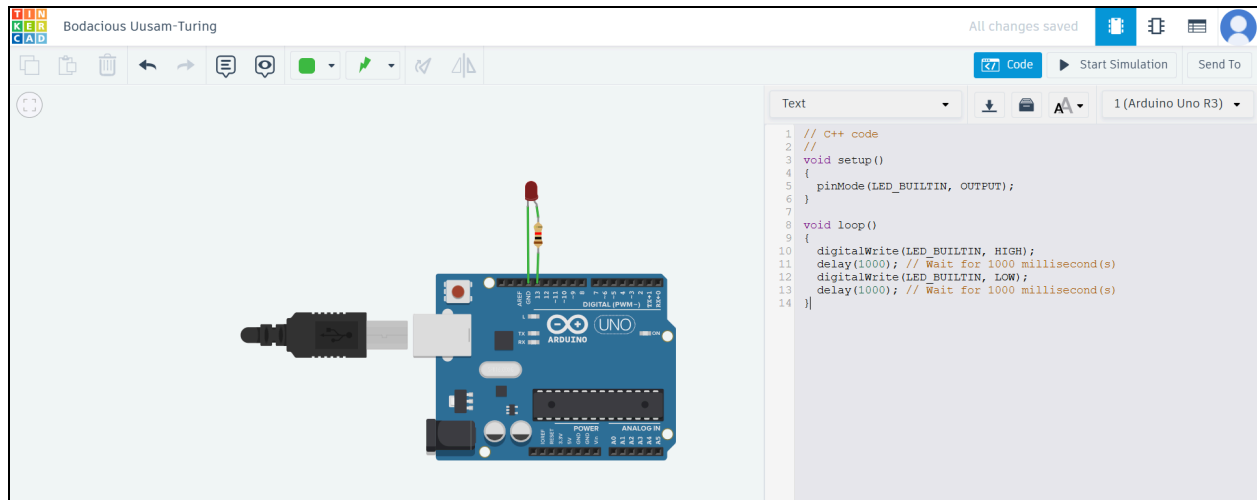




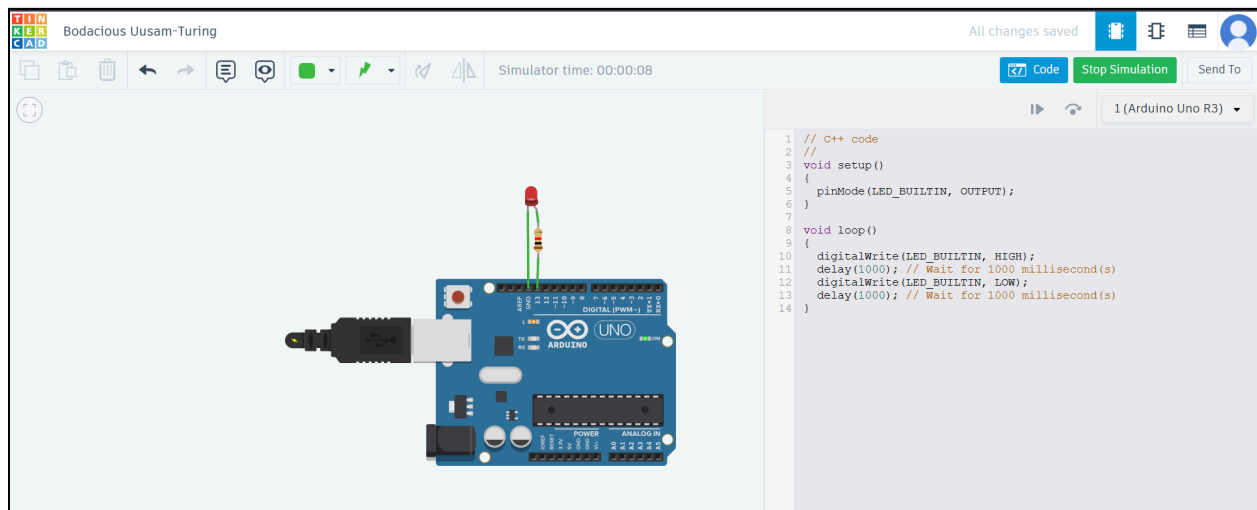
Click on the arduino and then click on 'code', present on the top right corner



From the dropdown, select 'Text' and enter your code



Click on the 'Start Simulation' button. The LED starts blinking every 1000ms



Click on the stop simulation button, to stop the simulation.

Conclusion:

In this simulation, we created a Temperature-Based Notice Board using Arduino and the TMP36 temperature sensor. The system directly reads temperature data from the sensor and displays it on an LCD in real-time. Additionally, we included hardcoded text in the code to display a welcome message or any other relevant information on the LCD when the system starts. Although the HC-05 Bluetooth module was not available for simulation in Tinkercad, we successfully completed the temperature sensor portion of the project. This simulation demonstrates how the system can display both real-time temperature readings and static messages, making it ideal for applications such as environmental monitoring, weather stations, or as a temperature display in classrooms or labs. The next step would be to integrate Bluetooth functionality to allow remote messaging.